

Sequence & Series

* Bounded Sequence

(i) Bounded below $\rightarrow$ lower term known but upper term unknown.

(ii) Bounded above $\rightarrow$ upper limit known $\rightarrow$

$$S_n = -n^2 \quad \forall n \in \mathbb{N}$$

$$\langle S_n \rangle = \{-1^2, -2^2, -3^2, \dots\}$$

$-1 < -1, -4 < -1$ upper limit

(iii) Bounded above & Bounded below $\rightarrow$ called Bounded Sequence.

$$\langle S_n \rangle = \langle 1/n \rangle$$

$$= \{1, \frac{1}{2}, \frac{1}{3}, \dots, 0\}$$

* Convergent:

$\lim_{n \rightarrow \infty} U_n = \text{finite (unique)}$

$$u = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$$

$$U_n = \frac{n}{n+1}$$

$$\lim_{n \rightarrow \infty} U_n = \frac{1}{1+1/n} = 1$$

* divergent

$\lim_{n \rightarrow \infty} U_n = \text{infinite}$

$$u = 1, 2, 3, \dots$$

$$U_n = n$$

$$\lim_{n \rightarrow \infty} U_n = \lim_{n \rightarrow \infty} n = \infty$$

* Oscillatory $\rightarrow$ not unique

if $U_n =$ not unique
 $n \rightarrow \infty$

$U_n \rightarrow 1, -1, 1, -1, \dots$

$\otimes S_n = 0$ if $n \rightarrow \text{even}$

$S_n = -1$ if $n \rightarrow \text{odd}$

Theorem: (a) The Geometric Series $1 + r + r^2 + \dots + r^n$

(i) Convergent - if $|r| < 1$

(ii) Divergent - if $|r| > 1$

(iii) Oscillatory if $r \leq -1$

* Comparison test

if $\sum U_n$ & $\sum V_n$ be two series of +ve terms

(a) $\sum V_n$ is convergent then $\sum U_n$ is convergent -
if $U_n \leq V_n$
if $\lim_{n \rightarrow \infty} \frac{U_n}{V_n} = k$ (finite +ve)

(b) $\sum V_n$ is divergent then $\sum U_n$ is divergent
if $\lim_{n \rightarrow \infty} \frac{U_n}{V_n} = k$ (finite +ve)

p-series test

the series $\sum \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \frac{1}{3^p} + \dots$

(i) Convergent - if $p > 1$

(ii) Divergent - if $p \leq 1$

$$e_m = 1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots + \frac{1}{2m-1} + \dots$$

Solⁿ. $U_m = \frac{1}{2m-1}$

Let $V_m = \frac{1}{m}$

Let $\lim_{m \rightarrow \infty} \frac{U_m}{V_m} = \frac{m}{2m-1} = \frac{1}{2 - 1/n} = \frac{1}{2}$ (finite +ve)

But $\sum V_m = \sum \frac{1}{m}$, here $p=1$ (By p-series test)
then $\sum V_m$ is divergent.

By comparison test U_m is divergent.

$$e_2 = \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \dots$$

Solⁿ. $U_m = \frac{2m-1}{m(m+1)(m+2)} = \frac{2m-1}{m^3 + 3m^2 + 2m}$

Let $V_m = \frac{1}{m^2}$

Let $\lim_{m \rightarrow \infty} \frac{U_m}{V_m} = \frac{m^2(2m-1)}{m^3 + 3m^2 + 2m}$
 $= \frac{\cancel{m^3} (2 - 1/m)}{\cancel{m^3} (1 + 3/m + 2/m^2)}$

$= \frac{2}{1} = 2$ (finite +ve)

But $\sum V_m = \sum \frac{1}{m^2}$, here $p=2$ (By p-test)
then V_m is convergent.

By comparison test U_m is convergent.

$$S_n = \sum_{m=1}^n \left(\sqrt{m^4+1} - \sqrt{m^4-1} \right)$$

$$\begin{aligned} \text{Sol}^n \quad V_n &= \sqrt{n^4+1} - \sqrt{n^4-1} \\ &= \frac{n^4+1 - (n^4-1)}{\sqrt{n^4+1} + \sqrt{n^4-1}} = \frac{2}{\sqrt{n^4+1} + \sqrt{n^4-1}} \end{aligned}$$

$$\text{Let } V_n = \frac{1}{n^2}$$

$$\begin{aligned} \text{Let } \lim_{n \rightarrow \infty} \frac{V_n}{V_n} &= \lim_{n \rightarrow \infty} \frac{2n^2}{\sqrt{n^4+1} + \sqrt{n^4-1}} \\ &= \frac{2}{\sqrt{1+\frac{1}{n^4}} + \sqrt{1-\frac{1}{n^4}}} \\ &= \frac{2}{1+1} = 1 \quad (\text{finite, } +ve) \end{aligned}$$

$$\sum V_n = \sum \frac{1}{n^2}, \quad p=2 \quad (\text{By } p\text{-series test})$$

$\sum V_n$ is convergent.

$\therefore$ by comparison test its convergent.

$$S_n = \frac{1}{n} \sin \frac{1}{n}$$

$$\text{Sol}^n \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$V_n = \frac{1}{n} \left(\frac{1}{n} - \frac{(1/n)^3}{3!} + \frac{(1/n)^5}{5!} - \dots \right)$$

$$\text{Let } V_n = \frac{1}{n^2}$$

$$\begin{aligned} \text{Let } \lim_{n \rightarrow \infty} \frac{V_n}{V_n} &= \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n^2 3!} + \frac{1}{n^2 5!} - \dots \right) \\ &= 1 \quad (\text{finite } +ve) \end{aligned}$$

$\sum V_n \rightarrow$ is convergent by $p=2$ (p -series)

$\therefore$ its convergent.

* Cauchy's Integral test

If for $n \geq 1$, $f(n)$ be a positive monotonic decreasing function

s.t. $f(n) = u_n$ for positive integral value of n , then

$\sum_{n=1}^{\infty} u_n$ is convergent if improper integral $\int_1^{\infty} f(x) dx$ is convergent otherwise $\sum u_n$ is divergent.

Ex. $1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

$$u_n = \frac{1}{n^2} = f(n)$$

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{x^2} dx = -\frac{1}{x} \Big|_1^{\infty} = -\left(0 - 1\right) = 1 \text{ finite}$$

$\therefore \int_1^{\infty} f(x) dx$ is finite $\rightarrow$ convergent

By Cauchy's test $\sum u_n$ is convergent

Ex. $\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots$

$$u_n = \frac{1}{(n+1)(n+2)} = f(n)$$

$$f(n) = \frac{1}{(n+1)(n+2)}$$

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} \left(\frac{1}{x+1} - \frac{1}{x+2} \right) dx = \left(\ln \frac{x+1}{x+2} \right) \Big|_1^{\infty}$$

$$\therefore \int_1^{\infty} f(x) dx \rightarrow \text{convergence} = \left(\ln \frac{1+\frac{1}{n}}{1+\frac{2}{n}} \right) \Big|_1^{\infty}$$

By Cauchy's test $\sum u_n$ is convergent $= 0 - \left(\ln \frac{2}{3} \right)$

$= \ln \frac{3}{2}$ — finite

Ex. Show that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ & divergent if $0 < p \leq 1$

$$u_n = \frac{1}{n^p} \Rightarrow f(n) = \frac{1}{n^p}$$

$$\int_1^{\infty} f(x) dx = \int_1^{\infty} \frac{1}{x^p} dx = \left[\frac{x^{-p+1}}{-p+1} \right]_1^{\infty} = \frac{1}{1-p} \Big|_1^{\infty} \text{ if } p \neq 1$$

$$= \left(\ln x \right) \Big|_1^{\infty} \text{ if } p = 1$$

$$\int_1^{\infty} \frac{1}{x^p} dx = \begin{cases} \alpha & p < 1 \\ \frac{1}{1-p} \left|_1^{\infty} \right. & p < 1 \\ \frac{1}{1-p} \left|_1^{\infty} \right. & p > 1 \end{cases} = \begin{cases} \alpha & p < 1 \\ \alpha & p < 1 \\ \frac{1}{1-p} \left|_1^{\infty} \right. & p > 1 \end{cases}$$

$$= \begin{cases} \alpha & p < 1 \\ (0 - \frac{1}{1-p}) & p > 1 \end{cases} = \begin{cases} \alpha & p < 1 \\ -\frac{1}{p-1} & p > 1 \end{cases}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \Rightarrow \begin{cases} \text{convergent if } p > 1 \\ \text{divergent if } p \leq 1 \end{cases}$$

* D'Alembert Ratio test:

The series $\sum u_n$ of positive term is -

(i) Convergent if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} < 1$

(ii) divergent if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} > 1$

(iii) Test fail if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1$

Concept:

$\lim_{n \rightarrow \infty} u_n = 0$ (convergent)

$\lim_{n \rightarrow \infty} u_n > 0$ (divergent)

Ex. $x^2 + 2x + 3x^2 + \dots$ if $x > 0 \rightarrow +ve$

$$u_n = n^2 x^{n-1}$$

$$u_{n+1} = (n+1)^2 (x^n) \quad \left| \quad \frac{u_{n+1}}{u_n} = \frac{(n+1)^2 x^n}{n^2 x^{n-1}} \right.$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^2 x^n}{n^2 x^{n-1}} = \lim_{n \rightarrow \infty} \frac{(1 + \frac{1}{n})^2 x}{1} = x$$

By D'Alembert ratio test - $\sum u_n$ is convergent if $x < 1$ & divergent if $x > 1$

If $x = 1$, $u_n = n^2$

$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} n^2 = \infty \leftarrow \text{divergent}$

Ex. $\frac{x}{1 \cdot 2} + \frac{x^2}{2 \cdot 3} + \dots$

$$u_n = \frac{x^n}{(2n-1)2n}$$

$$u_{n+1} = \frac{x^{n+1}}{(2n+1)(2n+2)}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{x^{n+1} (2n-1) 2n}{(2n+1)(2n+2) x^n}$$

$$= \lim_{n \rightarrow \infty} \frac{x (2n-1) 2n}{2 (2n+1)(n+1)}$$

$$= \lim_{n \rightarrow \infty} \frac{x (2 - \frac{1}{n}) 2}{2 (2 + \frac{1}{n})(1 + \frac{1}{n})}$$

$$= \frac{x \cdot 4}{2 \cdot 2 \cdot 1} = x$$

By D'Alembert's test

$\sum u_n$ is convergent if $x < 1$

- divergent if $x > 1$

If $x = 1$

$$u_n = \frac{1}{(2n-1)2n}$$

$v_n = \frac{1}{n^2}$ — convergent —

$$\lim_{n \rightarrow \infty} \frac{v_n}{u_n} = \frac{\frac{1}{n^2}}{\frac{1}{(2n-1)2n}}$$

$$= \lim_{n \rightarrow \infty} (2 - \frac{1}{n}) 2$$

= 4 — finite (≠ 0)

Convergent —

$\therefore \sum u_n$ is convergent — if $x \leq 1$

Ex. $1 + \frac{x}{2} + \frac{x^2}{5} + \dots + \frac{x^n}{n^2+1}$

$$u_n = \frac{x^n}{n^2+1}$$

$$u_{n+1} = \frac{x^{n+1}}{(n+1)^2+1}$$

$$\frac{u_{n+1}}{u_n} = \frac{(x^{n+1})(n^2+1)}{((n+1)^2+1)x^n}$$

$$= \frac{n^2+1}{n^2+2n+2} x$$

$$= \frac{1 + \frac{1}{n^2}}{1 + \frac{2}{n} + \frac{2}{n^2}} x$$

$\therefore x < 1 \rightarrow$ convergent —

$x > 1 \rightarrow$ divergent —

If $x = 1$

$$u_n = \frac{1}{n^2+1}$$

$$v_n = \frac{1}{n^2}$$

$$\frac{v_n}{u_n} = 1 + \frac{1}{n^2}$$

$\lim_{n \rightarrow \infty} \frac{v_n}{u_n} = 1 \rightarrow$ finite (≠ 0) $\sum u_n \rightarrow$ convergent

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} x = x$$

* Cauchy's Root test:

A series $\sum u_n$ of positive term if

(i) convergence if $\lim_{n \rightarrow \infty} (u_n)^{1/n} < 1$

(ii) divergence if $\lim_{n \rightarrow \infty} (u_n)^{1/n} > 1$

(iii) Test fail if $\lim_{n \rightarrow \infty} (u_n)^{1/n} = 1$

Ex. $\sum_{n=1}^{\infty} \left[\frac{\left(\frac{n+1}{n}\right)^{n^2}}{3^n} \right] = u_n$

$$(u_n)^{1/n} = \left(\frac{\left(\frac{n+1}{n}\right)^{n^2}}{3^n} \right)^{1/n}$$

$$\lim_{n \rightarrow \infty} (u_n)^{1/n} = \lim_{n \rightarrow \infty} \left[\frac{\left(\frac{n+1}{n}\right)^n}{3} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)^n}{3} = \frac{e}{3} \rightarrow \text{finite } (< 1)$$

$\therefore$ By Cauchy's root test - $\sum u_n \rightarrow$ convergent -

Ex. $u_n = \frac{n^{n^2}}{(n+1)^{n^2}}$

$$\lim_{n \rightarrow \infty} (u_n)^{1/n} = \lim_{n \rightarrow \infty} \frac{n^n}{(1+n)^n}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e} < 1$$

$\therefore$ By Cauchy's root test - u_n is convergent -

Ex. $u_n = \frac{n^3}{3^n}$

$$\lim_{n \rightarrow \infty} (u_n)^{1/n} = \lim_{n \rightarrow \infty} \frac{n^{3/n}}{3} = \lim_{n \rightarrow \infty} \frac{\left(n^{1/n}\right)^3}{3} = \frac{1}{3} < 1$$

$\therefore$ By Cauchy's root test series is convergent -

* Raabe's test:

The series $\sum u_n$ of positive term is

(i) convergence if $\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) > 1$

(ii) divergence if $\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) < 1$

(iii) Test fail if $\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = 1$

Ex. $1 + \frac{3}{7}x + \frac{3 \cdot 6}{7 \cdot 10}x^2 + \frac{3 \cdot 6 \cdot 9}{7 \cdot 10 \cdot 13}x^3 + \dots$

$$u_n = \frac{3 \cdot 6 \cdot 9 \dots (3n-3)}{7 \cdot 10 \cdot 13 \dots (3n+1)} x^n$$

$$u_{n+1} = \frac{3 \cdot 6 \cdot 9 \dots 3n}{7 \cdot 10 \cdot 13 \dots (3n+4)} x^{n+1}$$

$$\frac{u_{n+1}}{u_n} = \frac{3n}{3n+4} x$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{3}{3+\frac{4}{n}} x = x$$

$\therefore$ By ratio test - Series is

i) convergent if $x < 1$

ii) divergent if $x > 1$

(iii) Test fail if $x = 1$

$\therefore$ Use at $x = 1$

Raabe's test - $\frac{u_{n+1}}{u_n} = \frac{3n}{3n+4}$

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{3n+4}{3n} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} n \left(\frac{3n+4-3n}{3n} \right) = \frac{4}{3} > 1$$

$\therefore$ By Raabe's test - Series is convergent at $x = 1$.

$$\text{Ex. } n^4 + \frac{2^2}{3 \cdot 4} n^4 + \frac{2^2 \cdot 4^2}{3 \cdot 4 \cdot 5 \cdot 6} n^6$$

$$u_n = \frac{2^2 \cdot 4^2 \cdot 6^2 \cdots (2n)^2}{3 \cdot 4 \cdot 5 \cdot 6 \cdots (2n+1)(2n+2)} n^{2n+2}$$

$$u_{n+1} = \frac{(2n+2)^2}{(2n+3)(2n+4)} n^{2(n+1)+2}$$

$$\frac{u_{n+1}}{u_n} = \frac{(2n+2)^2}{(2n+3)(2n+4)} n^2$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{4}{2 \cdot 2} n^2 = n^2$$

By ratio test - Series is

i) Convergent if $n^2 < 1$

ii) Divergent if $n^2 > 1$

iii) Test fail if $n^2 = 1$

Let $n^2 = 1$

$$\frac{u_{n+1}}{u_n} = \frac{(2n+2)^2}{(2n+3)(2n+4)}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) &= \lim_{n \rightarrow \infty} n \left(\frac{(n^2 + 8n)}{1n^2 + 8n + 4} \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{6 + 8/n}{1 + 8/n + 4/n^2} \right) = 6 > 1 \end{aligned}$$

By Raabe's test - Series is convergent at $n^2 < 1$

$$\text{Ex. } 1 + \frac{a(a+1)}{1 \cdot 2} + \frac{a(a+1)(a+2)}{1 \cdot 2 \cdot 3} \cdots$$

$$u_n = \frac{a(a+1)(a+2) \cdots (a+(n-1))}{1 \cdot 2 \cdot 3 \cdots n}$$

$$u_{n+1} = \frac{(a+n)}{n+1}$$

$$\frac{u_{n+1}}{u_n} = \frac{a+b}{n+1}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = 1$$

Ratio test fail

By Raabe's test -

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \lim_{n \rightarrow \infty} n \left(\frac{n+1}{a+b} - 1 \right)$$

$$= \lim_{n \rightarrow \infty} n \left(\frac{n+1 - a - b}{a+b} \right)$$

$$\lim_{n \rightarrow \infty} \left(\frac{1-a}{1+a/b} \right) = 1-a$$

By Raabe's test -

i) Convergent if $1-a > 1$

ii) Divergent if $1-a < 1$

iii) If $1-a = 1 \Rightarrow a = 0$

then Series = 1 ← Convergent

* Alternating Series:

A series of the type $u_1 - u_2 + u_3 - u_4 + \dots$, where $u_n > 0 \forall n \in \mathbb{N}$ is called alternating series & denoted by

$$\sum (-1)^{n-1} u_n$$

$$S_n = \sum (-1)^{n-1} = 1 - 1 + 1 - 1 - \dots$$

$$\sum (-1)^{n-1} \frac{1}{\sqrt{n}} = 1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$$

→ Leibnitz Test for the convergence of an alternating series -

If an alternating series

$$\sum_{n=1}^{\infty} (-1)^{n-1} u_n = u_1 - u_2 + u_3 - u_4 + \dots$$

is satisfying following conditions

(i) Each term is numerically less than the preceding term i.e. $u_{n+1} < u_n \forall n \in \mathbb{N}$

$$(ii) \lim_{n \rightarrow \infty} u_n = 0$$

Then the series $\sum (-1)^{n-1} u_n$ is convergent.

$$\text{Ex. } 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots$$

$$(i) u_{n+1} < u_n \forall n \in \mathbb{N}$$

$$(ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n!} = 0$$

∴ by Leibnitz test the series is convergent.

$$S_n = \frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{3}-1} + \frac{1}{\sqrt{4}-1} - \dots$$

$$(i) u_{n+1} < u_n \forall n \in \mathbb{N}$$

$$(ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}-1} = 0$$

∴ By Leibnitz test the series is convergent.

Ex. Prove that the series $\sum_{n=0}^{\infty} (-1)^n \frac{x^n}{n!}$ is

(i) convergent - if $|x| < 1$ (ii) divergent - if $|x| > 1$

$$- u_n = \frac{(-1)^n x^n}{n!} \quad \vee \quad u_{n+1} = \frac{(-1)^{n+1} x^{n+1}}{(n+1)!}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| &= \lim_{n \rightarrow \infty} \left| \frac{n}{n+1} x \right| \\ &= |x| \lim_{n \rightarrow \infty} \left| \frac{1}{1 + \frac{1}{n}} \right| = |x| \end{aligned}$$

Hence, by D'Alembert's test, the given Series
convergent - if $|x| < 1$ & divergent - if $|x| > 1$.
& test fail - if $|x| = 1$

Now

$$x = 1$$

$$1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots$$

$$(i) u_{n+1} < u_n \quad \forall n \in \mathbb{N}$$

$$(ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

By Leibnitz test the series
is convergent at $x = 1$

$$x = -1$$

$$1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \dots$$

$$1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \dots$$

$$u_n = \frac{1}{n^p} \quad \forall p = 1$$

By p-series test -
if $p = 1 \rightarrow$ divergent -

* Absolute convergent -

A series $\sum u_n$ which contains both the positive & negative terms is said to be absolute convergent -
if the series $\sum |u_n|$ is convergent -

Ex. $\sum u_n = 1 - \frac{1}{2} + \frac{1}{2^2} - \frac{1}{2^3} + \dots$

$$(i) u_{n+1} < u_n \quad \forall n \in \mathbb{N}$$

$$(ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$$

$\therefore$ By Leibnitz test, the series is convergent -

$$\sum |u_n| = 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{n-1}}$$

$$r = \frac{1}{2} < 1$$

$\therefore \sum |u_n|$ is convergent -

Hence, Given Series is absolutely convergent -

Ex. Show that the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n^2}$

= $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$ is an absolute convergent

$$\therefore (i) u_{n+1} < u_n \quad \forall n \in \mathbb{N}$$

$$(ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} (-1)^{n+1} \frac{1}{n^2} = 0$$

By Leibnitz test the series is convergent -

Now. $\sum |u_n| = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots + \frac{1}{n^2}$

By p-series test, the series is convergent as $p = 2$ ($p > 1$)

Hence, Series $\sum (-1)^{n+1} \frac{1}{n^2}$ is absolutely convergent.

Ex. $1 - \frac{n^2}{2!} + \frac{n^4}{4!} - \dots$

$$u_n = \frac{(-1)^n n^{2n}}{2n!} \quad u_{n+1} = \frac{(-1)^{n+1} n^{2n+2}}{(2n+2)!}$$

$$(i) u_n > u_{n+1} \quad \forall n \in \mathbb{N}$$

$$(ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{(-1)^n n^{2n}}{2n!} = 0$$

$\therefore$ By Leibnitz test series is convergent -

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2n! n^2}{(2n+2)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{1}{(2n+2)(2n+1)} \right|$$

$$= 0 < 1$$

$|(-1)^{n+1} u_n|$ is convergent -

$\therefore$ the series is absolutely convergent

→ Conditionally convergent:

→ A series $\sum u_n$ is said to be conditionally convergent or semi convergent - if $\sum u_n$ is convergent - but it is not absolutely convergent -

Ex . $u_n = 1 - \frac{1}{2} + \frac{1}{3} - \dots$

i) $u_n > u_{n+1} \quad \forall n \in \mathbb{N}$

ii) $\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$

By Leibnitz test series is convergent -

$$\sum |(-1)^n u_n| = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

$$u_n = \frac{1}{n}, \quad p=1 \leftarrow \text{divergent by } p\text{-series test}$$

$\therefore \sum (-1)^n u_n$ is conditionally convergent -